

Project Design Phase

Solution Architecture

Date	11 September 2026
Team ID	SWUID2026136950
Project Name	Automated Network Request Management System (ServiceNow)
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Architecture Overview

The Network Request Management solution is built entirely on the ServiceNow Cloud Platform as a layered architecture. Four user personas — Requester, Approver, Network Engineer, and Admin — interact with the system through the browser, email, or the ServiceNow UI. Their actions flow through five internal layers:

- **Presentation Layer** — the Service Portal (HTML5, CSS3, AngularJS) hosts the Catalog Item form, with dynamic UI Policies controlling which fields appear based on the selected request type, and giving requesters live status tracking.
- **Automation Layer** — Flow Designer drives the end-to-end process: requesting approval, creating the Catalog Task, waiting on conditions, and updating records — with no manual handoffs.
- **Business Logic Layer** — Business Rules, Client Scripts, UI Policies, and ACLs enforce defaults, validation, and access control across every transaction.

- **Data Layer** — the custom network_request table, along with sc_req_item, sc_task, sysapproval_approver, sys_user, and sys_user_group, plus the CMDB for device reference, together hold the full record of every request.
- **Notification Layer** — an SMTP-based email engine sends the four lifecycle notifications (submission, approval request, decision, completion) to keep every stakeholder informed in real time.

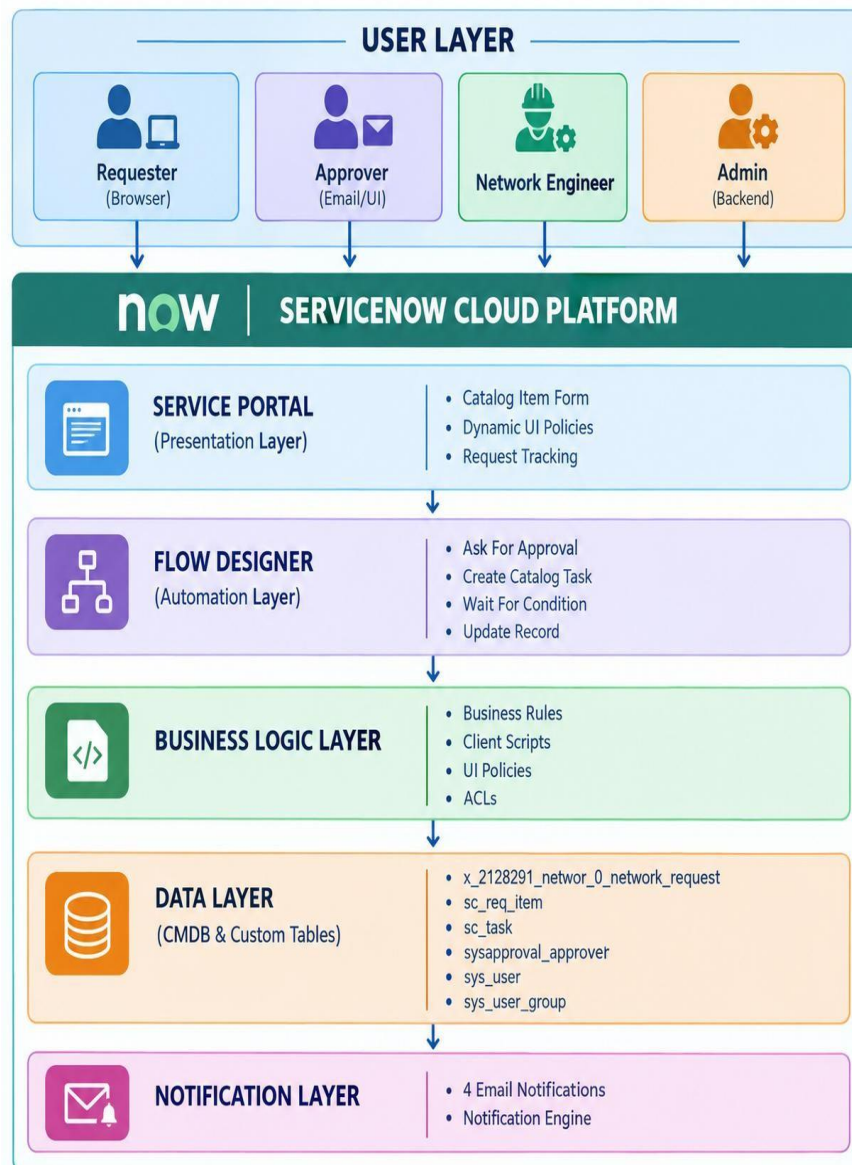


Figure 1: Layered solution architecture of the Network Request Management system on the ServiceNow Cloud Platform.

End-to-End Process Flow: The diagram below traces a single request through all nine steps of the system, from submission to reporting, showing the actor, the action taken, and the ServiceNow tool responsible at each stage.



Figure 2: Step-by-step process flow —from request submission through approval, fulfillment, and reporting.

Technology Stack

Each architectural layer maps to specific ServiceNow components and technologies, summarized below.

Layer	Component	Technology	Purpose
Presentation	Service Portal	HTML5, CSS3, AngularJS	User interface for request submission

Presentation	UI Policies	ServiceNow Client-side	Dynamic field visibility
Automation	Flow Designer	ServiceNow Flow Engine	End-to-end workflow automation
Layer	Component	Technology	Purpose
Automation	Business Rules	JavaScript (server-side)	Default values, auto-assignment
Security	ACLs	ServiceNow Security	Record-level access control
Data	Custom Table (x_2128291_networ_0_network_request)	ServiceNow Database	Store network request data
Data	CMDB	ServiceNow CMDB	Device reference
Data	sc_req_item	ServiceNow Database	Requested Item records
Data	sc_task	ServiceNow Database	Catalog Task records
Notification	Email Engine	SMTP	Send 4 lifecycle emails
Reporting	Reports & Dashboard	ServiceNow Analytics	Metrics and SLA tracking